

LLM Model Based Approach For Data Extraction

Gemini 2.5 Pro model

Gemini 2.5 Pro supports input in the form of images and provides output in a properly structured format based on the extracted information

Price of the model:

Tier	Input Price	Output Price (including thinking tokens)
Free Tier	Free of charge	Free of charge
Paid Tier	\$1.25 per 1M tokens (prompts $\leq$ 200k tokens), \$2.50 per 1M tokens (prompts $>$ 200k tokens)	\$10.00 per 1M tokens (prompts $\leq$ 200k tokens)

Free Tier

- **Input price:** Free → you can send text to the model without paying.
- **Output price:** Free → the model's generated text is also free.
- **Token limits:**
 - Input limit: 1,048,576 tokens → that's the maximum you can send in a single request.
 - Output limit: 65,536 tokens → the maximum the model can generate in one response.

Paid Tier

- **Input price:** You pay based on how many tokens you send:
 - \$1.25 per 1M tokens if your prompt is $\leq$ **200k tokens**
 - \$2.50 per 1M tokens if your prompt is $>$ **200k tokens**
- **Output price:** You pay \$10 per 1M tokens generated (including "thinking tokens").

A) \$1.25 per 1M tokens if your prompt is $\leq$ 200k tokens

Parameter	Value
Input token cost	\$1.25 per 1M tokens = 0.00000125 USD/token
Output token cost	\$10.00 per 1M tokens = 0.00001 USD/token
Input tokens per image	1500
Output tokens per image	300
Input cost per image	$1500 \times 0.00000125 = 0.001875$ USD
Output cost per image	$300 \times 0.00001 = 0.003$ USD
Total cost per image	$0.001875 + 0.003 = 0.004875$ USD
Total cost for 1000 images	$1000 \times 0.004875 = 4.875$ USD

B) \$2.50 per 1M tokens if your prompt is $>$ 200k tokens

Parameter	Value
Input token cost	\$2.50 per 1M tokens = 0.0000025 USD/token
Output token cost	\$15.00 per 1M tokens = 0.000015 USD/token
Input tokens per image	1500
Output tokens per image	300
Input cost per image	$1500 \times 0.0000025 = 0.00375$ USD
Output cost per image	$300 \times 0.000015 = 0.0045$ USD
Total cost per image	$0.00375 + 0.0045 = 0.00825$ USD
Total cost for 1000 images	$1000 \times 0.00825 = 8.25$ USD

Google Vision AI Billing & OCR Overview

Free Tier

- Google Cloud provides \$300 in free credits for new users.
- Billing applies once free credits are used or for usage beyond the free tier.

Document Text Detection (OCR) Pricing

Number of Images/Month	Cost
First 1,000 images	FREE
1,001 – 5,000,000	\$1.50
5,000,001+	\$0.60

Note: After OCR, the extracted text can be further processed to generate structured data.

Google Vision OCR Cost (Paid Tier)

From Google Vision OCR paid pricing:

- \$1.50 per 1000 images per month
- \$0.0015 per image

Self-Hosted Model:

Ollama Model: LLaMA 3.2 Vision

The LLaMA 3.2 Vision model in Ollama is a multimodal large language model that supports both text and image inputs. It is designed for tasks that require understanding and reasoning over text and images simultaneously. LLaMA 3.2 Vision is free to use with Ollama when run locally but on the other hand, these locally run models take a long time to process for image data extraction compared to other models.

Key Features:

1. Multimodal Input:

- Accepts both text and image as input, enabling applications like document analysis, visual question answering, and image-based data extraction.

2. Long Context Support:

- Can handle very long inputs of up to 128,000 tokens in a single request.
- This allows the model to process large documents or multiple combined inputs without losing context.

3. Single Input Processing:

Optimized to process a single input at a time, which can include both textual and visual data.

4. The output response is up to 4k tokens.

OpenAI–GPT MODELS

A) GPT-4.1 Model

Parameter	Value
Input token cost	\$2 per 1M tokens = 0.000002 USD/token
Output token cost	\$8 per 1M tokens = 0.000008 USD/token
Input tokens per image	1500
Output tokens per image	300
Input cost per image	$1500 \times 0.000002 = 0.003$ USD
Output cost per image	$300 \times 0.000008 = 0.0024$ USD
Total cost per image	$0.003 + 0.0024 = 0.0054$ USD
Total cost for 1000 images	$1000 \times 0.0054 = 5.4$ USD

B) GPT-4o mini

Parameter	Value
Input token cost	\$0.15 per 1M tokens = 0.00000015 USD/token
Output token cost	\$0.60 per 1,000,000 tokens = 0.00000060 USD/token
Input tokens per image	1500
Output tokens per image	300
Input cost per image	$1500 \times 0.00000015 = 0.000225$ USD
Output cost per image	$300 \times 0.00000060 = 0.00018$ USD
Total cost per image	$0.000225 + 0.00018 = 0.000405$ USD
Total cost for 1000 images	$1000 \times 0.000405 = 0.405$ USD

Claude Model

Claude Haiku 3

Parameter	Value
Input token cost	\$0.25 per 1M tokens = 0.00000025 USD/token
Output token cost	\$1.25 per 1M tokens = 0.00000125 USD/token
Input tokens per image	1500
Output tokens per image	300
Input cost per image	$1500 \times 0.00000025 = 0.000375$ USD
Output cost per image	$300 \times 0.00000125 = 0.000375$ USD
Total cost per image	$0.000375 + 0.000375 = 0.00075$ USD
Total cost for 1000 images	$1000 \times 0.00075 = 0.75$ USD

Amazon Textract

Region:Asia (Mumbai) – ap-south-1

Number of Images/Month	Cost
First 1,000 images	FREE/Per month
First million pages in a month	\$1.50
Over 1 million pages in a month	\$0.60

AWS Textract cost (after first 1,000 free)

- **Cost per image:** 0.0015 USD
- **Cost for 1,000 images:** $1,000 \times 0.0015 = \mathbf{1.50 \text{ USD}}$

Combined cost

Number of Images	Claude Haiku 3	AWS Textract	Total Cost
1	0.00075 USD	0.0015 USD	0.00225 USD
1000	0.75 USD	1.50 USD	2.25 USD